

ORGANIC CHEMISTRY – ALKANES & COMBUSTION

KEY VOCABULARY

Alkane: saturated hydrocarbon, C_nH_{2n+2} .

Saturated: only single C–C bonds.

Complete combustion: burning in plenty of oxygen $\rightarrow CO_2 + H_2O$.

Incomplete combustion: limited oxygen $\rightarrow CO$ and/or carbon (soot).

COMMON MISCONCEPTIONS

~~Alkanes have a C=C double bond.~~

✓ Alkanes only have single bonds (saturated).

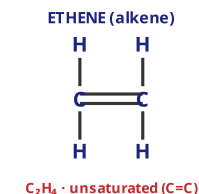
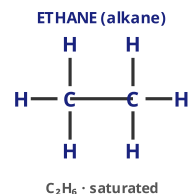
~~Complete combustion makes carbon monoxide.~~

✓ Complete combustion makes carbon dioxide and water.

~~Soot is produced with plenty of oxygen.~~

✓ Soot forms in incomplete combustion (limited O_2).

ETHANE (AN ALKANE)



1 GENERAL FORMULA

Give the **general formula** of an alkane. [1 mark]

2 COMPLETE COMBUSTION

Write the products of the **complete combustion** of methane. [2 marks]

3 INCOMPLETE COMBUSTION

Explain why incomplete combustion is dangerous. [2 marks]

4 BALANCE COMBUSTION

Complete: $CH_4 + __ O_2 \rightarrow CO_2 + __ H_2O$. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Complete combustion of a hydrocarbon makes _____ and _____.

carbon dioxide

carbon monoxide

water

soot

6 WHY OXIDATION

Explain why combustion is an oxidation reaction. [1 mark]